

Imaginary quadratic fields

Prime generating polynomials, complex multiplication, $e^{\sqrt{-5}}$

1. Prime generating polynomials

Lemma: Let R be a Noetherian domain. Then any element can be written as the product of irreducibles.

Corollary: A Noetherian domain R is a UFD iff all irreducibles are prime.

Prop: Let $N: \mathbb{C} \rightarrow \mathbb{C}$ denote the norm, $\mathbb{Z} \subset R \subset \mathbb{C}$ a Noetherian domain. If for any $n, m \in \mathbb{R}^*$ there are $\lambda, \mu \in R$ s.t. $0 < N(\lambda n + \mu m) < N(n)$ then R is a UFD.

Proof: If R is not a UFD, take an irreducible $q \in R$ which is not prime. Let $a, b \in R$ be s.t. $p \nmid a, b, p \nmid ab$ and $N(ab) = \min Q$, where $Q = \{N(a'b'): p \nmid a', b', p \nmid ab'\}$. Let $\lambda, \mu \in R$ be s.t. $0 < N(\lambda q + \mu a) < N(a)$. Then $(\lambda q + \mu a)b \in Q$, so

$$N(ab) \leq N(\lambda q + \mu a)b = N(\lambda q + \mu a)N(b) < N(a)N(b) = N(ab),$$

a contradiction. $\square$

Note: In a Euclidean domain, the Euclidean algorithm produces $\lambda=1, \mu=-q, r=n-qn$.

Def: We call $(u, v) \in (R^*)^2$ s.t. such λ, μ do not exist a **bugging pair**, and $\frac{u}{v}$ a **bugging fraction**, i.e. a fraction for which $0 < N(\lambda \frac{u}{v} + \mu) < 1$ has no solutions in R .

Lemma: Let $\frac{u}{v}$ be a bugging fraction. Then elements of $\frac{u}{v} + R, (\frac{u}{v} + \mathbb{Z}) \setminus \mathbb{Z}$

are also bugging.

Prop: If R is not a UFD then there is a bugging pair $(p-\theta, q) \in R^2$ with $p, q \in \mathbb{Z}$, q prime in $\mathbb{Z}$ and θ is not divisible in R by an integer.

Proof: We may assume $\frac{u}{v} = \frac{p+R\theta}{Q}$ with $P, R, Q \in \mathbb{Z}$ coprime. By multiplying, we may also assume $Q = q$ is prime in $\mathbb{Z}$.

• R, q are coprime:

Let d be their highest common factor. Then

$$\frac{P+R\theta}{q} \cdot \frac{q}{d} = \frac{P}{d} + \frac{R}{d}\theta.$$

If $d \neq 1$ then $\frac{P}{d} \in \mathbb{Q} \setminus \mathbb{Z}$ so that this is also a bugging fraction. Since $\frac{P}{d} \in \mathbb{Z}$, so is $\frac{P+RQ}{q} \cdot \frac{q}{d} - \frac{P}{d}\theta = \frac{P}{d} \in \mathbb{Q}$. But this is impossible.

Let $x, y \in \mathbb{Z}$ be s.t. $Rx - qy = 1$. Then $\frac{P+RQ}{q} x - y\theta = \frac{Px - \theta}{q}$

is also a bugging fraction. Now let $Px = zq + p$, using the Euclidean algorithm.

Then $\frac{Px - \theta}{q} - z = \frac{p - \theta}{q}$ is a bugging fraction. $\square$

Now let $R = \mathbb{Z}[\theta]$ where $\theta = \frac{1 + \sqrt{-4m}}{2}, m \in \mathbb{Z}$. If $\frac{p-\theta}{q}$ is bugging then

$$N(\frac{p-\theta}{q}) = \frac{p^2 - p + m}{q^2} \geq 1,$$

i.e. $p^2 - p + m \geq q^2$. Since $p < q$, we also get $q < m$. Hence, for such R , we have

Prop: Let $R = \mathbb{Z}[\theta]$ of the form above. If R is not a UFD then there is a bugging fraction of the form $\frac{p-\theta}{q}$ with $p < q < m$ and q prime in $\mathbb{Z}$.

Thm: If $p^2 - p + m \in \mathbb{Z}$ is prime for $p = 1, \dots, m-2$ then R is a UFD.

Proof: Let $p < q < m$. If $p^2 - p + m$ is prime then it is equal or coprime to q . In the first case $\frac{p-\theta}{q}$ is not bugging since $N(\frac{p-\theta}{q}) = \frac{1}{q} < 1$. In the second case $\exists x, y \in \mathbb{Z}$ with

$$(p^2 - p + m)x - qy = 1.$$

Hence, $\frac{p-\theta}{q} \cdot (p-\theta)x - y = \frac{1}{q}$. Taking norms, we see that $\frac{p-\theta}{q}$ cannot be bugging. $\square$

Lemma: If $p < m$ then $p-\theta$ is irreducible.

Proof: If $p-\theta = xy$ for $x, y \in R$ non-units, then $N(xy) = p^2 - p + m < m^2$, so x or y has a norm $< m$. Now, $x, y \in \mathbb{Z}[\theta] \setminus \mathbb{Z}$. We show that for such elements, the norms are always larger than m :

Let $a + b\theta \in \mathbb{Z}[\theta] \setminus \mathbb{Z}$. Then

$$N(a + b\theta) = a^2 + ab + mb^2 \stackrel{\alpha\beta > \alpha + \beta}{\geq} a^2 + ab + m + b^2 - 1 \stackrel{\geq m}{\geq} m.$$

$$\stackrel{\geq m}{\geq} m. \quad \alpha^2 + \alpha\beta + \beta^2 \geq 1 \text{ for } x, y \in \mathbb{Z} \setminus 0 \quad \square$$

Thm: If R is a UFD then $p^2 - p + m$ is prime for $p = 1, \dots, m-1$.

Proof: Assume $p^2 - p + m$ is not prime. Then

$$(p-\theta)(p-\bar{\theta}) = p^2 - p + m = q_1 \dots q_\ell$$

for $\ell > 1$. Even if q_i is not prime inside R , this gives non-unique factorisations. $\square$

Note: From both theorems together, it follows that if $x^2 - x + m$ is prime for $x = 1, \dots, m-2$ then so is $(m-1)^2 - (m-1) + m = m^2 - 2m + 2$.

2. Complex multiplication

Prop: Let $\Gamma = \text{PSL}_2(\mathbb{Z})$. Call two lattices $\Lambda, \Lambda' \subset \mathbb{C}$ homothetic if $\exists z \in \mathbb{C}$ s.t. $\Lambda' = z\Lambda$. Then lattices in $\mathbb{C} / \sim_{\text{homothety}} \cong \mathbb{H} / \Gamma \xrightarrow{\sim} \mathbb{P}^1$.

Prop: Let E be an elliptic curve. Then $E \cong \mathbb{C} / \Lambda$ for some lattice $\Lambda \subset \mathbb{C}$.

Def: Let E be an elliptic curve. Then E has **complex multiplication** if $\mathbb{Z} \neq \text{End} E \subset \mathbb{C}$.

Prop: If E has CM then $\text{End} E \otimes \mathbb{Q}$ is an imaginary quadratic field and $\text{End} E \subset \text{End} E \otimes \mathbb{Q}$ is a lattice.

Let K be an imaginary quadratic field with ring of integers R_K . Let $I \subset K$ be a fractional ideal, i.e. an R_K -submodule s.t. $\exists r \in R_K \setminus 0$ with $rI \subset R_K$.

Then $I \subset K$ is a lattice. Let E_I denote the associated elliptic curve. Then

$$\text{End} E_I \cong \{ \alpha \in \mathbb{C} : \alpha I \subset I \}$$

$$= \{ \alpha \in K : \alpha I \subset I \}$$

$$= R_K.$$

Hence, $\text{Cl}(R_K) = \text{fractional ideals} / \text{principal ideals} \rightarrow \text{Ell}(R_K) = \text{ell. curves with } \text{End} E \leq R_K / \sim$.

Prop: Let Λ be a lattice with $E_\Lambda \in \text{Ell}(R_K)$, $I, J \subset K$ fractional ideals. Then

(i) $I\Lambda \subset \mathbb{C}$ is a lattice;

(ii) $\text{End}(E_{I\Lambda}) \cong R_{K, I}$

(iii) $E_{I\Lambda} \cong E_{J\Lambda}$ iff $I = J$ in $\text{Cl}(R_K)$;

In particular, $\text{Cl}(R_K) \cong \text{Ell}(R_K)$ by $I E_\Lambda = E_{I\Lambda}$.

(iv) $\text{Cl}(R_K) \cong \text{Ell}(R_K)$ is simply transitive.

$\swarrow$ This is actually an equality, and $= [K]_{\mathbb{C}} \cong \mathbb{H} / \Gamma_K$

Prop: Let E have CM. Then $[\mathbb{Q}[j(E)]: \mathbb{Q}] \leq [\text{Cl}(R_K)]$.

Proof: Let $\sigma \in \text{Aut} \mathbb{C}$. Then $j(E^\sigma) = j(E)^\sigma$, as j is a rational function of the coefficients in Weierstrass form. Since $\text{End} E^\sigma \cong \text{End} E \leq R_K$, $E^\sigma \in \text{Ell}(R_K)$.

Hence, $|G_j(E)| \leq |\text{Ell}(R_K)| = |\text{Cl}(R_K)|$. $\square$

Prop: Let E have CM with $\text{End} E = R_K$. Then $j(E)$ is an algebraic integer.

Proof: Note that E has CM iff it has an endomorphism of non-square degree. We will study the set of elliptic curves with an isogeny $E \rightarrow E'$ and later take $E' = E$.

Let $n \in \mathbb{Z}$, $D_n = \{ \begin{pmatrix} a & b \\ 0 & b \end{pmatrix} \in \text{Mat}_2(\mathbb{Z}) : ad - bc = n \}$,

$$S_n = \{ \begin{pmatrix} a & b \\ 0 & b \end{pmatrix} \in D_n : 0 \leq b < \sqrt{n} \}.$$

Note that $S_n = D_n / \text{SL}_2(\mathbb{Z})$. For $\alpha \in \text{Mat}_2(\mathbb{Z})^0$, let

$$F_n(X) = \prod_{\alpha \in S_n} (X - j \circ \alpha) = \sum_m s_m X^m, \quad \uparrow \text{symmetric functions of } j \circ \alpha$$

(Claim 1: $s_m \circ \gamma = s_m \quad \forall \gamma \in \text{SL}_2(\mathbb{Z})$)

Let $S_n \in \text{SL}_2(\mathbb{Z})$ be s.t. $S_n(\alpha\gamma) \in S_n$. Then the map $S_n \rightarrow S_n$, $\alpha \mapsto S_n \alpha \gamma$

is injective, hence bijective. Now

$$\{ j \circ (\alpha\gamma) : \alpha \in S_n \} = \{ j \circ S_n^{-1} \circ S_n \alpha \gamma : \alpha \in S_n \} = \{ j \circ S_n \alpha \gamma : \alpha \in S_n \} \text{ since } j \text{ is } \text{SL}_2(\mathbb{Z})\text{-invariant} = \{ j \circ \alpha : \alpha \in S_n \}.$$

Hence, any symmetric function on this set is invariant under $\gamma \mapsto \gamma\gamma$.

Claim 2: $s_m \in \mathbb{C}[j]$

We only need to show that s_m is meromorphic at $i\infty$, since the algebra of modular functions is $\mathbb{C}(j)$ and s_m is holomorphic.

If $\alpha = \begin{pmatrix} a & b \\ 0 & b \end{pmatrix} \in S_n$ then $j \circ \alpha(\tau) = e^{-2\pi i \frac{a\tau+b}{b}} + \sum_{k \neq 0} c_k e^{2\pi i k \frac{a\tau+b}{b}}$.

In particular, $q^{n+1} j \circ \alpha(\tau) \rightarrow 0$ as $q \rightarrow 0$. Hence, $\exists N > 0$ s.t. $q^N s_m(\tau) \rightarrow 0$ as $q \rightarrow 0$.

Claim 3: s_m has Fourier coefficients in $\mathbb{Z}$

We have $q^n \alpha(\tau) = e^{2\pi i \frac{a\tau+b}{b}} = q^{\frac{a}{b}} Q^{\frac{a\tau}{b}}, \quad q^{\frac{a}{b}} = q^{\frac{a}{b}}$

$$j \circ \alpha(\tau) = q^{\frac{a}{b}} Q^{-\frac{a}{b}} + \sum_{k \neq 0} c_k q^{\frac{a}{b} + \frac{ak}{b}} Q^{\frac{ak}{b}}.$$

Hence, s_m also has Fourier coefficients in $\mathbb{Z}[\frac{1}{b}]$. Let $\sigma \in \text{Gal}(\mathbb{Q}(\sqrt{n})/\mathbb{Q})$, and $\sqrt{n}^\sigma = \sqrt{n}^{(r)}$ for some $r(\sigma) \in \mathbb{Z}$. Then, coefficient-wise,

$$(j \circ \alpha)^\sigma = \sqrt{n}^{r(\sigma)\frac{a}{b}} Q^{-\frac{a}{b}} + \sum_{k \neq 0} c_k \sqrt{n}^{r(\sigma)\frac{a}{b} + \frac{ak}{b}} Q^{\frac{ak}{b}},$$

i.e. $(j \circ \alpha)^\sigma = j \circ \begin{pmatrix} a & r(\sigma)b \\ 0 & b \end{pmatrix}$. Note that $j \circ \begin{pmatrix} a & b \\ 0 & b \end{pmatrix}$ only depends on $b \pmod{a}$, by multiplying by $\begin{pmatrix} r & n \\ 0 & b \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$. Since $(r, n) = 1$, $\{ r \pmod{a} : 0 \leq b < \sqrt{n} \} = \mathbb{Z}/a\mathbb{Z}$, and hence $\{ j \circ \begin{pmatrix} a & r(\sigma)b \\ 0 & b \end{pmatrix} : \begin{pmatrix} a & b \\ 0 & b \end{pmatrix} \in S_n \} = \{ j \circ \alpha : \alpha \in S_n \}$.

Since s_m is a symmetric function on this set, this implies that its Fourier coefficients are in $\mathbb{Q}$, and hence in $\mathbb{Z} = \mathbb{Z}[\sqrt{n}] \cap \mathbb{Q}$.

Claim 4: $s_m \in \mathbb{Z}[j]$

We will show that $\mathbb{C}[j] \cap \mathbb{Z}[[q, q^{-1}]] = \mathbb{Z}[j]$. Let

$$f(j) = a_0 j^3 + \dots + a_N \in \mathbb{C}[j] \cap \mathbb{Z}[[q, q^{-1}]].$$

Then $f(q) = \frac{a_0}{q^3} + \frac{a_1 + 744}{q^{2+1}} + \dots$. Hence, $a_0 \in \mathbb{Z}$. Repeat with $f - a_0 j^3$.

These claims show that $F_n(Y, X) \in \mathbb{Z}[Y, X]$, where $\prod_{\alpha \in S_n} (X - j \circ \alpha) = F_n(j, X)$.

Claim 5: Let $\beta \in D_n$. Then $j \circ \beta$ is integral over $\mathbb{Z}[j]$

Let $\gamma \in \text{SL}_2(\mathbb{Z})$ be s.t. $\gamma\beta \in S_n$. Then $j \circ \beta = j \circ \gamma\beta$, which is a root of $F_n(j, X)$.

Claim 6: If n is not a square then $H_n = F_n(X, X) \in \mathbb{Z}[X] \setminus \mathbb{Z}$ and has leading coefficient $\neq 1$.

For $\alpha \in S_n$, we have

$$j \circ \alpha = \left(\frac{1}{Q^n} + \sum_{k \neq 0} c_k Q^{nk} \right) - \left(\frac{1}{q^{ab} Q^{a^2}} + \sum_{k \neq 0} c_k q^{abk} Q^{a^2k} \right).$$

Since n is non-square, $j \circ \alpha$ has a pole at $Q=0$ and the leading coefficient is a root of unity. Hence, the same is true about $F_n(j, j)$, which has integer coefficients. Hence, its leading coefficient is $\neq 1$ and

$$F_n(j, j) = \pm q^{-m} + \dots \in q^{-m} \mathbb{Z}[[q]]$$

for some $m > 0$. Hence, $F_n(j, j) = \pm j^m + \dots \in \mathbb{Z}[j]$.

Claim: $j(E)$ is an algebraic integer

Let $\text{End} E \leq R_K \ni q$ s.t. $N(q) = n$ is not a square. Then $q: E \rightarrow E$ has degree n . Let $j(E) = j(\tau)$ for $\tau \in \mathbb{H}$. Then $[\mathbb{Z} + \tau\mathbb{Z} : q(\mathbb{Z} + \tau\mathbb{Z})] = n$.

Let $q(\tau) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \tau \\ 1 \end{pmatrix} = \alpha \begin{pmatrix} \tau \\ 1 \end{pmatrix}$,

for $\alpha \in D_n$. Since $q \in \text{End} E$,

$$j(\alpha\tau) = j(\tau) = j(E),$$

so

$$0 = F_n(j, j \circ \alpha)(\tau) = F_n(j(E), j(E)) \quad \square$$

3. $e^{\sqrt{-163}}$

Prop: Let K be a number field. Then R_K is a UFD iff $\text{Cl}(R_K) = 1$.

Prop: Let $K = \mathbb{Q}[\sqrt{D}]$. Then

$$R_K \cong \begin{cases} \mathbb{Z}[\sqrt{D}] & \text{if } D \not\equiv 1 \pmod{4} \\ \mathbb{Z}[\frac{1+\sqrt{D}}{2}] & \text{if } D \equiv 1 \pmod{4} \end{cases}$$

Proof: Clearly, $\mathbb{Z}[\sqrt{D}] \subset R_K$. Let $a + b\sqrt{D} \in K \setminus \mathbb{Z}[\sqrt{D}]$. Its minimal polynomial over $\mathbb{Q}$ is $x^2 - 2ax + (a^2 - Db^2)$.

So $a + b\sqrt{D} \in R_K$ iff $2a, a^2 - Db^2 \in \mathbb{Z}$. If $a \in \mathbb{Z}$ then also $b \in \mathbb{Z}$. So assume $a = \frac{\alpha_0}{2} \in \frac{1}{2}\mathbb{Z} \setminus \mathbb{Z}$. Then $a^2 - Db^2 = \frac{\alpha_0^2 - (2b)^2 D}{4}$.

Since $\alpha_0^2 \equiv 1 \pmod{4}$, this is in $\mathbb{Z}$ iff $(2b)^2 D \equiv 1 \pmod{4}$. In particular, $2b \in \mathbb{Z}$ but $b \notin \mathbb{Z}$, i.e. $b \in \frac{1}{2}\mathbb{Z} \setminus \mathbb{Z}$. Hence, $2a, a^2 - Db^2 \in \mathbb{Z}$ iff either $a, b \in \mathbb{Z}$ or $a, b \in \frac{1}{2}\mathbb{Z} \setminus \mathbb{Z}$ and $D \equiv 1 \pmod{4}$. $\square$

We can check, using chapter 1, that the following fields have rings of integers which are UFDs:

$$\mathbb{Q}[\sqrt{-1}], \mathbb{Q}[\sqrt{-2}], \mathbb{Q}[\sqrt{-3}], \mathbb{Q}[\sqrt{-7}], \mathbb{Q}[\sqrt{-11}], \mathbb{Q}[\sqrt{-19}],$$

$$\mathbb{Q}[\sqrt{-43}], \mathbb{Q}[\sqrt{-67}], \mathbb{Q}[\sqrt{-163}].$$

It turns out that this is a complete list of imaginary quadratic fields for which this is true.

Now consider $R_{\mathbb{Q}[\sqrt{-163}]} = \mathbb{Z}[\frac{1+\sqrt{-163}}{2}]$. By results from chapter 2, we have

$$[\mathbb{Q}[\frac{1+\sqrt{-163}}{2}]: \mathbb{Q}] = h_K = 1,$$

i.e. $j(\frac{1+\sqrt{-163}}{2}) \in \mathbb{Q}$, and that $j(\frac{1+\sqrt{-163}}{2})$ is an algebraic integer. Hence,

$$j(\frac{1+\sqrt{-163}}{2}) \in \mathbb{Z}.$$

In general, for $q = e^{2\pi i \tau}$, $j(q) = \frac{1}{q} + 744 + 196884q + O(q^2)$.

Hence, $j(\frac{1+\sqrt{-163}}{2}) = e^{-\pi i (1+\sqrt{-163})} + 744 + 196884 e^{-\pi i (1+\sqrt{-163})} + O(e^{-2\pi (1+\sqrt{-163})})$

$$= -e^{\pi \sqrt{-163}} + 744 - \underbrace{196884 e^{\pi \sqrt{-163}} + O(e^{-2\pi \sqrt{-163}})}_{\ll 1}$$

$$\approx -e^{\pi \sqrt{-163}} + 744.$$

In particular, $e^{\pi \sqrt{-163}}$ is almost an integer.